

Voice-Light: A Full-Duplex Cascaded Voice Agent with Causal Turn-Taking and Speculative Generation

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Abstract

Natural spoken interaction requires more than streaming ASR, language generation, and speech synthesis: a system must react to overlap without canceling on every acknowledgment, prepare a response before a turn is certain, and ensure canceled audio cannot enter conversation history. We present VOICE-LIGHT, a full-duplex cascaded voice agent that combines immediate acoustic onset, a causal adapter sharing a streaming ASR encoder, reversible playback control, and private speculative response generation. Structured tool calls execute concurrently with audible bridge speech, while browser acknowledgments make rendered audio authoritative for durable history. Locked evaluation on 1,673 real-conversation silence candidates found that an earlier learned completion checkpoint preserved a 2.70% false-cutoff rate but reached only 12.53% end-of-turn recall, compared with 95.60% for a Silero timing policy. The deployed system therefore retains a hybrid controller rather than claiming a learned-policy replacement. Across three unscripted operator-run microphone sessions, 36 measured response turns had a 758 ms median from final VAD endpoint to first server audio; 21 turns were below 800 ms. These sessions are an instrumented case study, not a controlled user evaluation. We release the synthetic data, model artifacts, evaluation code and summaries, source code, and deployment configuration supporting the result.

Keywords: streaming voice agents, turn-taking, synthetic data, causal streaming inference, speech synthesis, tool use, deployment

1 Introduction

A conventional voice assistant is often drawn as a serial pipeline: automatic speech recognition (ASR), then a language model, then text-to-speech (TTS). That abstraction hides the interaction problem. A usable conversational system must decide whether a silence is a completed turn or a within-turn hesitation; react immediately when speech begins during playback without canceling on every short acknowledgment; ensure canceled audio cannot reappear; and distinguish generated text from speech that actually reached the listener.

Here, *full duplex* means that microphone ingestion continues during assistant playback, playback can react reversibly to detected user speech, and generation, synthesis, and acknowledged audio are managed concurrently.

VOICE-LIGHT retains a cascaded architecture because its boundaries expose typed tools, playback state, and cancellation. Its central design rule is that uncertain work may begin early, but it may become audible or durable only after explicit causal checks. This rule connects three otherwise separate problems: turn inference must not inspect future audio, speculative generation must remain private until promotion, and conversation history must contain only speech acknowledged by the listener’s browser.

The study asks three questions. First, can a small causal adapter reuse features from a persistent streaming ASR encoder and improve turn commitment over deployable timing baselines? Second, can a reversible controller exploit uncertain evidence without turning every VAD onset into cancellation? Third, how much response latency can private speculation hide in a deployed ASR–LLM–TTS cascade?

The paper makes four scoped contributions:

- a shared-encoder causal turn-taking adapter, together with locked evaluations that report both the learned signal and its failure to replace stronger timing baselines;
- a typed hybrid controller for reversible ducking, floor-taking

interruption, backchannel resumption, stale-generation rejection, and audible-only history;

- private speculative generation integrated with streaming ASR, Qwen, Kyutai TTS, and sequential tool rounds in a public scale-to-zero deployment; and
- released synthetic tool-use and turn-taking corpora, trained artifacts, evaluation code and result summaries, and an instrumented 36-turn latency case study.

The contributions are systems and evaluation contributions. The paper does not claim that the adapter is a new state of the art, that synthetic evaluation measures general tool competence, or that the microphone sessions estimate population-level interaction quality.

2 Related work

Turn-taking and endpointing. TurnGPT predicts turn completion from text [3]; Voice Activity Projection (VAP) predicts future speaker activity directly from conversational audio [4]; and acoustic–language-model fusion shows that lexical and prosodic evidence can be complementary [5]. Recent turn benchmarks further emphasize that endpointing, backchannels, and interruption vary with conversation type [6, 7]. VOICE-LIGHT does not propose a general turn-taking architecture. It asks whether a small causal adapter can share the encoder already used for streaming ASR and whether its evidence remains useful after a locked comparison with complete deployable policies.

Full-duplex spoken agents. End-to-end systems such as Moshi jointly model user and assistant speech and avoid explicit ASR–LLM–TTS boundaries [8]. A cascade instead exposes intermediate text, typed tools, playback state, and component failures. VOICE-LIGHT uses those boundaries to make overlap reversible, reject stale work, and construct history

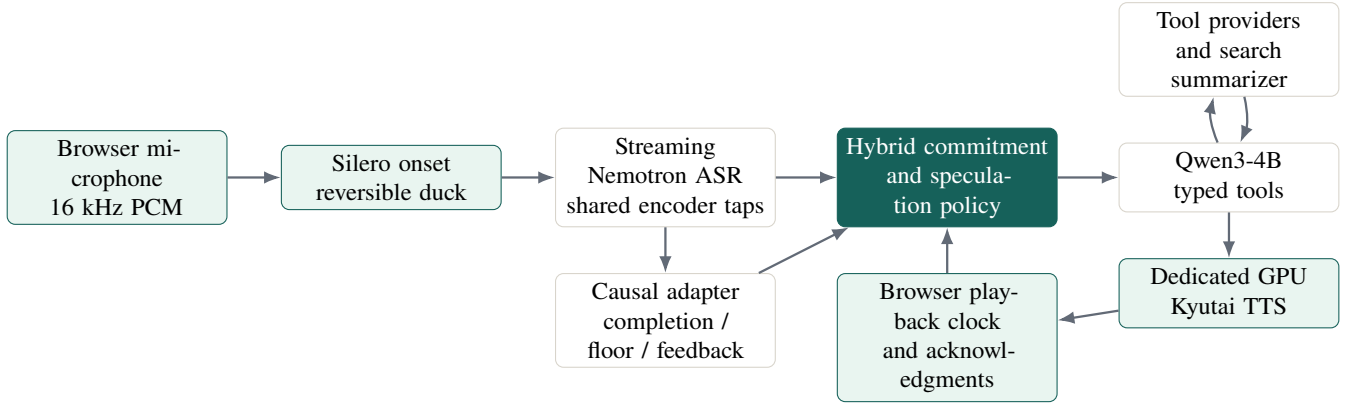


Figure 1: Production runtime. Immediate acoustic reaction and learned interaction evidence meet at a typed controller. Tool calls and results use a structured loop; acknowledged browser playback is authoritative for durable assistant history.

from acknowledged audio. Its latency results are system measurements and are not directly comparable with model-internal latency reported for end-to-end architectures. Speculative end-turn detection has also combined local and remote detectors [9]; here, *speculation* means privately preparing downstream text and PCM before a final turn commitment.

Synthetic tool use. Toolformer and ToolLLM construct supervision for API selection and multi-step execution [10, 11]. The Voice-Light tool branch addresses a narrower spoken protocol: short audible bridges, causally ordered call/result records, sequential calls, and suppression of protocol markup. Its shared-generator holdout measures acquisition of that protocol, not general tool use or factual retrieval.

3 System overview

Figure 1 shows the deployed data path. The browser streams 16 kHz microphone PCM over one WebSocket and acknowledges rendered audio ranges. Silero provides the earliest acoustic onset signal. A persistent Nemotron worker produces streaming transcripts and exposes selected encoder layers to the causal adapter without loading a second speech backbone. The controller combines those signals with transcript revisions, playback state, and conservative deadlines. Qwen generates speech and typed calls; Kyutai TTS [17] synthesizes only text released by the controller.

Four invariants define the system. All learned turn features are causal and share the ASR stream. Speech onset may duck or pause playback immediately, but only stronger lexical, learned, or timeout evidence commits cancellation. Speculative text and PCM carry generation identifiers and remain private until transcript and turn checks promote them. Finally, generated assistant text becomes durable conversation context only to the extent that browser acknowledgments show it was audible.

4 Training data

The project produced two model-training branches. The language-model branch generated typed spoken-tool conversations for a Qwen adapter. The turn-taking branch combined synthetic interaction timelines with automatically prepared human conversations for a causal speech adapter. Figure 2 shows where the branches remain separate and where they meet the deployed system.

4.1 Synthetic spoken tool use

Spoken tool use was treated as a data and protocol problem rather than a prompt-only feature. A pinned Qwen3.6-27B-FP8 teacher generated provider-neutral records for natural conversation, drafting, brainstorming, search, calculation, time queries, sequential calls, and correction or recovery. The published corpus contains 3,999 conversations, 11,810 user messages, 15,308 assistant messages, and 3,498 structured calls across `search`, `calculate`, and `get_time`. It is available through the project collection on Hugging Face [27].

Generation used a causal conversation state machine. The teacher produced a user turn and then either assistant speech or speech followed by a typed call. The controller executed or synthesized the result, appended the linked call and result, and only then requested the next assistant step. Calculator and time results were deterministic. Synthetic search results came from the teacher, not the live web; consequently, the corpus teaches call ordering and grounding to supplied text, not factual retrieval quality.

The canonical JSONL is semantic rather than tokenized. It stores runtime tool definitions, audible assistant text, typed calls, stable identifiers, outcomes, later turns, split metadata, provenance, and hashes. The run pins the teacher revision, prompt revision, source commit, scenario seeds, quantization, and time anchor. Structural checks retained schema validity, causal call/result order, deterministic tool checks, spoken-length limits, and protocol-leak detection. Subjective teacher-based rejection was disabled after manual review found its false positives counterproductive.

4.2 Synthetic conversational turn-taking audio

The second generator models coherent conversation timelines rather than isolated completion utterances. Typed plans distinguish five semantic conditions: genuine user completion, a HOLD pause followed by continuation, non-floor feedback during assistant speech, a response after the assistant yields, and deliberate interruption. The plan also contains ordinary event-free user, assistant, and silence spans.

Only user audio is synthesized. Virtual assistant text keeps the scenario coherent and estimates assistant duration, but assistant waveform never becomes an input artifact. The runtime-relevant assistant-speaking probability is compiled as a causal state curve with ramps, dips, and bounded noise. This

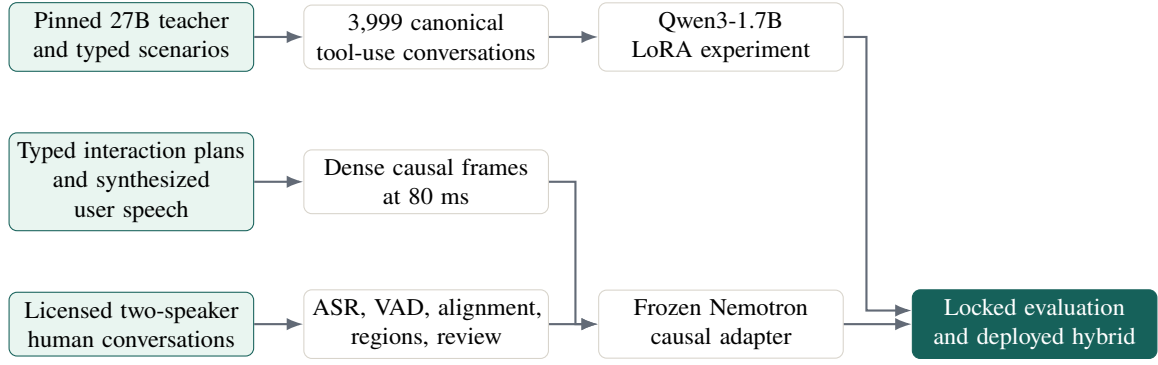


Figure 2: Two model-training branches feed one evaluated deployment. The turn-taking branch combines synthetic interaction timelines with separately prepared human evidence; the language-model branch uses synthetic typed conversations.

prevents future assistant state or clean synthetic timing from leaking into the adapter.

The accepted rendering route used Qwen VoiceDesign references to condition CosyVoice 3 zero-shot voices [19, 18]. Each user unit was rendered and trimmed independently; the compiler then measured its active speech, assembled the final timeline, and rasterized 20-second views into 250 frames at 80 ms. A planned pause became HOLD supervision only if the rendered unit contained at least 500 ms of silence followed by resumed speech.

The public synthetic repository [30] stores lossless speech units and reconstruction metadata rather than duplicating every composed waveform. Plans, voice provenance, trim measurements, composition seeds, split assignments, and checksums allow conversation audio and training crops to be rebuilt. The retained V4 and V5 manifests contain 1,092 and 1,097 timelines, approximately 21.84 and 21.80 hours of source timeline respectively. These counts describe retained run manifests, not a claim that synthetic interaction validation substitutes for natural conversation.

4.3 Automatic preparation of human conversations

The human pipeline separates three artifact layers: authorized source audio, recording-level evidence, and materialized training windows. Each complete speaker track is stored once as lossless FLAC. Typed Parquet rows reference bounded 20-second intervals and contain aligned inputs, targets, and masks; they do not duplicate the recording.

Ingestion registers and hashes both channels, reuses exact cached transcripts, transports only missing material, applies a reviewed inter-speaker offset, and persists a full-duration annotation and quality result transactionally. Multiple ASR and activity systems provide independent evidence for consensus alignment, channel-aware crosstalk filtering, conversation regions, and quality flags. Supplied TurnBench annotations remain an independent evidence source rather than being copied into model-output fields to manufacture agreement.

The completed preparation pass accepted eight MagicHub conversations (about 2.77 hours) [20] and 37 Mundo TurnBench conversations (about 7.31 hours) [6]; a silent TurnBench recording was excluded. These additions joined previously prepared authorized conversational material. The locked corpus manifest records 107 accepted conversations totaling 36.3 hours across conversation-disjoint train, validation, and test splits. A manual pipeline-validation audit reviewed 12 dataset-stratified recordings and applied targeted exclusions; broader

Table 1: Human conversational material in the locked corpus manifest. Source audio remains subject to its original access terms; only derived manifests and code are distributed by the project.

Material	Conversations	Hours
MagicHub	8	2.77
TurnBench	37	7.31
Additional authorized material	62	26.22
Total	107	36.30

corpus quality was tracked through the automated evidence pipeline.

5 Model adaptation

5.1 A language adapter for spoken tools

The tool-use experiment fine-tuned Qwen/Qwen3-1.7B [13] with a BF16 rank-16 LoRA [12]. Two source records from each combination of eight behavior families and five speech styles formed a balanced 80-record validation split; the remaining 3,919 records were training data. Across 16 deterministic passes, every record appeared once per pass. Intact short segments were regrouped into histories with 8–16 user turns, producing 14,391 training histories and 17.84 million rendered tokens, of which 6.86 million were assistant targets.

The native Qwen chat template rendered the canonical records with thinking disabled. The loss mask retained ordinary assistant speech, the spoken bridge preceding a call, structured call tokens, post-result continuation, and later assistant turns. System text, tool definitions, user messages, tool results, padding, and non-assistant protocol tokens were masked. The adapter targeted all attention and MLP projections, trained 17.43 million parameters, and completed 904 optimizer steps in 47 minutes on an RTX 4090.

The public deployment uses Qwen3-4B Instruct [14]; the trained 1.7B LoRA remains a released experimental artifact [28, 29]. Section 6 reports why the synthetic protocol result was not treated as evidence of general conversational quality.

5.2 A causal adapter on shared Nemotron features

The turn-taking model attaches approximately 183,000 trainable parameters to the frozen Nemotron Speech Streaming 0.6B encoder [16]. Nemotron is a cache-aware streaming FastConformer-RNNT. Its activation caches allow new non-overlapping chunks to reuse prior work [1, 2]. This property

made it possible to share one streaming backbone between ASR and turn inference rather than re-encoding a rolling waveform or loading a second 0.6-billion-parameter model.

The adapter taps encoder layers 6, 12, 18, and 24. Each 1,024-dimensional tap is normalized and projected to 32 dimensions; the streams are fused to 64 dimensions, processed by residual causal depthwise-separable convolutions, combined with assistant-speaking state, and passed through a single-layer 64-dimensional unidirectional GRU. Separate heads emit completion/yield evidence, interaction events including floor take and non-floor feedback, and future activity. Convolution and recurrent state persist incrementally and reset with the ASR stream.

Synthetic pretraining selected step 2,250. Human fine-tuning warm-started those weights with a fresh optimizer, 15% synthetic replay, and 1,884 human boundaries: 1,038 HOLD and 846 end-of-turn (EOT) examples. The resulting 750-step checkpoint was selected on human validation.

The final adapter-best.pt is step 750, with the pinned Nemotron revision listed in Table 6, one lookahead token, 80 ms encoder frames, and the four taps above. Human validation selected it; synthetic validation could not. At runtime, forward hooks capture the exact features produced by persistent streaming ASR. A bounded latest-value queue supersedes stale adapter work, and loading or inference failure degrades the predictor without stopping ASR or the voice session.

The offline protocols evaluate several checkpoints with different roles. Section 6 keeps those roles explicit rather than transferring results between artifacts.

6 Offline evaluation

6.1 Protocols and checkpoint lineage

The project uses separate evaluation protocols for separate claims. Tool-use evaluation measures behavior on held-out synthetic conversation states. Turn completion has two non-comparable protocols: the original V1 benchmark scores future silence under a yield-oriented target and opens a locked test split only after policy selection on validation, whereas V2 scores semantic completion at speech boundaries. V2 stopped at validation because no detector met its predefined deployment gate, so its test split remained sealed. Deployment evidence is an operator-run microphone case study, not a substitute for a controlled conversational study.

Table 2: Turn-adapter checkpoint lineage. Parenthesized values are training steps. “Validation” and “test” name distinct locked inventories; evidence is not transferred between artifacts.

Checkpoint (step)	Role and evidence
Synthetic (2,250)	Pretraining seed; synthetic and human validation
Historical (3,500)	V1 locked test and V2 validation
Human fine-tune (750)	Deployed adapter; human validation and run-time only
Completion challenger (625)	Later experiment; V2 validation only

6.2 Synthetic tool-protocol results

The final adapter emitted 200/210 expected calls, 198/210 exact tool names, 200/210 schema-valid arguments, 198/210 concise bridges, 90/90 correct no-tool responses, and 180/180 valid

Table 3: Balanced synthetic holdout for Qwen3-1.7B. The same 160 states were decoded with three seeds. Expected calls, no-tool states, and continuations contribute 210, 90, and 180 repeated predictions respectively; percentages therefore do not represent independent tasks.

Behavior	Base	Fine-tuned
Correct tool decision	80.0%	95.2%
Exact tool name	78.6%	94.3%
Valid argument schema	80.0%	95.2%
Concise 1–12-word bridge	7.6%	94.3%
Correctly avoided a tool	70.0%	100.0%
Valid post-tool continuation	98.3%	100.0%

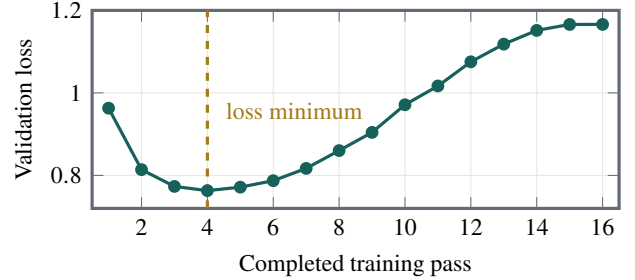


Figure 3: Recorded assistant-token validation loss after each logical pass. The directly comparable zero-update loss was 4.5398 and is omitted to preserve resolution. Loss bottomed after pass 4, then rose even while free-generation protocol metrics improved.

post-result continuations. These descriptive gains demonstrate acquisition of the generator’s conversational protocol. They do not establish factual accuracy, robust web search, or general tool intelligence because evaluation points share the training generator and schema.

6.3 Causal turn-completion results

Synthetic pretraining exposed the transfer problem directly. Completion AUROC was 0.9260 on 893 held-out synthetic examples but 0.5646 on 789 clean human validation labels. Human fine-tuning raised human AUROC to 0.5972. At its selected validation policy, the deployed step-750 checkpoint reached 65.04% EOT recall (426/655), 3.73% false cutoffs (5/134 HOLD cases), and 2.0 s p95 commitment latency. These are validation results, not locked-test results, and the head remained a poor standalone probability estimator.

For the policy metrics below, a HOLD candidate is a silence followed by continuation from the same speaker, while EOT denotes a completed user turn. False cutoff is the fraction of HOLD candidates committed as EOT; EOT recall counts completed turns committed before the timeout. Latency measures the causal delay from the candidate boundary to commitment, capped by the earlier of the timeout or the observed opportunity end.

The locked V1 test contains 1,673 causal silence candidates from 11 conversations, including only 37 HOLD cases. Threshold, minimum delay, and timeout were selected on validation and frozen before test. Table 4 shows the central negative result: the Voice-Light checkpoint preserved a low false-cutoff rate but crossed its learned threshold on only 205 of 1,636 EOT cases. Most turns therefore used the 800 ms timeout, and the learned policy did not beat the Silero or LiveKit timing baselines. This test evaluates historical step 3,500, not the deployed step-750 adapter.

The detectors do not score after identical amounts of silence. Voice-Light emits its boundary score at 80 ms, Smart Turn at 240 ms, LiveKit at 320 ms, and Silero updates a silence proxy on 32 ms chunks. Comparisons therefore evaluate complete causal policies at their native gates rather than isolating model quality under identical acoustic evidence. The implementations are pinned to Silero VAD 6.2.1, Pipecat Smart Turn v3.2, and LiveKit v1-mini [21, 22, 23].

Table 4: Locked real-conversation test under the original V1 yield-target protocol. Mean latency includes timeout actions. VL is historical Voice-Light step 3,500; policy values are threshold, minimum delay, and timeout. Possible training-source overlap makes Smart Turn contextual only.

Metric	VL	Silero	Smart	LiveKit
False cutoff	2.70%	2.70%	13.51%	2.70%
HOLD errors ($n = 37$)	1	1	5	1
EOT recall	12.53%	95.60%	20.72%	91.50%
Mean latency	770 ms	656 ms	684 ms	654 ms
Threshold	0.90	0.05	0.95	0.15
Minimum delay	560 ms	640 ms	80 ms	640 ms
Timeout	800 ms	800 ms	800 ms	800 ms

The V2 semantic-completion validation inventory contains 1,005 soft targets and 789 clean hard labels: 134 HOLD and 655 EOT examples, with 216 ambiguous cases excluded from hard scoring. Historical step 3,500 achieved AUROC 0.6866, 65.80% recall, 3.73% false cutoffs, and 2.0 s p95 latency. Its BCE of 0.6719 and Brier score of 0.1843 were worse than the constant soft-prior baseline (0.5847 and 0.1453), despite above-chance ranking. No swept detector met the joint gate of at most 5% false cutoff, at least 70% recall, and at most 800 ms p95 latency, so the V2 test split remained sealed. A completion-primary retrain improved calibration but still missed the gate; longer training reduced recall.

7 Streaming controller and deployment

7.1 Speculation and cancellation

After 80 ms of low Silero speech probability, the adapter receives another 80 ms opportunity to open a speculative response candidate. If learned evidence is not ready, a Silero fallback may start at 160 ms. Candidate text and PCM remain private until final ASR and transcript-revision checks permit promotion. Revisions limited to case, punctuation, whitespace, or apostrophes preserve work; lexical change invalidates it.

The system uses generation identifiers, cancellation barriers, and rebased sample positions to reject stale text or PCM. Playback clocks and boundary acknowledgements return every 80 ms. Durable assistant history is constructed only from audio ranges that the browser confirms it rendered; generated but canceled words remain diagnostic evidence rather than conversation content.

7.2 Tool execution and speech boundaries

Structured tools support calculation, current time, and web search, including sequential rounds. Tool execution can overlap with bridge audio already buffered in the browser. Kyutai, however, requires text lookahead. The bridge must therefore be closed before awaiting a tool result, and the result begins a second TTS utterance within the same assistant generation. Reusing one uninterrupted TTS session was implemented and reverted because it could strand the last bridge words until

tool completion, producing a deterministic mid-preamble stall. The final design avoids that stall but can retain a short acoustic seam at the semantic boundary. Before synthesis, a narrow text-boundary filter removes Markdown formatting, quotation marks, and emoji that Kyutai would otherwise pronounce poorly; the durable generated text remains unchanged.

A separate Qwen3-0.6B worker [15] summarizes web results so provider output does not occupy the main conversation worker. Tavily latency varied from approximately 234–408 ms in several calls to roughly 1.98–2.29 seconds in slower observations. External provider time is therefore reported separately from first-response latency.

7.3 Hybrid overlap policy

Silero remains the immediate acoustic authority. When user speech begins during assistant playback, the browser fades toward -15 dB over 450 ms and pauses by 500 ms. This response is reversible: onset alone does not discard generation. A floor-take probability above the configurable 0.82 threshold can commit interruption and cancellation. A non-floor-feedback probability above 0.82 becomes actionable after the short speech burst ends and resumes the same generation without adding a user turn. Explicit stop, repair, question, or meaningful lexical evidence provides another fast path.

Ambiguous overlap is bounded. Sustained speech commits floor taking after 900 ms, while a short ended burst resolves conservatively as non-floor feedback. Empty or slow final ASR receives a 120 ms grace rather than parking playback indefinitely. Generation and TTS budgets are held after 350 ms of unresolved overlap, and paused audio remains resumable for at most 800 ms. A committed interruption rejects new server PCM immediately and fades no more than 100 ms of already-buffered audio.

The adapter catches up from bounded pre-roll after Silero detects user activity, using the same persistent Nemotron stream once active. This preserves one backbone and strict causality while limiting the acoustic context available at the earliest overlap decision.

7.4 Two GPUs and scale to zero

Production requests a co-located pair of A10 GPUs, with an L40S pair as the capacity fallback. GPU 0 runs Nemotron, the adapter, Qwen3-4B, and the small search summarizer. GPU 1 is reserved for Kyutai. Moving Nemotron beside Kyutai improved Qwen first-delta latency by roughly 25–30 ms but regressed Kyutai first-word-to-PCM in small warm traces: 346–353 ms with dedicated TTS ($n = 3$) versus 459–486 ms in the alternate placement ($n = 5$). Because first audible speech dominates perceived response time, the dedicated-TTS topology was retained; the traces are engineering placement checks rather than statistical tests.

Modal [24] is a thin wrapper around the provider-neutral compute application. Model revisions and the adapter are pinned; Hugging Face and Torch caches live on a persistent volume; serving operates offline after initialization. Model subprocesses persist inside a warm container.

The final topology loads Nemotron, Qwen, Kyutai, and the search summarizer concurrently after the small VAD is ready. Cached weights prevent downloads but not image scheduling, imports, CUDA initialization, or framework startup. Observed cold readiness remains approximately 32–55 seconds. GPU

snapshots failed for the multi-process CUDA layout, and an import-only CPU snapshot added about 30 seconds. Keeping a container warm was rejected for the expected low visit frequency and cost.

8 End-to-end latency case study

The deployment case study comprises three unscripted microphone sessions run by one operator from a browser in Germany against the scale-to-zero service in Modal’s European region. Measurements begin after the service reported readiness and therefore exclude cold start. Prompts covered ordinary conversation, longer responses, interruptions, short acknowledgments, calculations, current information, and sequential tool use. The final VAD endpoint is the last speech boundary used for turn commitment; first server audio is the first PCM packet sent for the response.

Across 36 measured response turns, the median final-VAD-to-first-server-audio latency was 758 ms; 21/36 turns were below 800 ms, and the range was 528–1,652 ms. One session exported the complete telemetry trace for 13 turns. In that subset, median endpoint commitment was 502 ms, LLM first word 336 ms after generation start, TTS first PCM 444 ms after its first word, and server-to-browser render 95 ms. These component intervals overlap and must not be added.

Nine of the 13 fully traced turns promoted speculative work. Their median final-VAD-to-first-server-audio latency was 667 ms, compared with 1,513 ms for the four turns without promotion. This observational split is consistent with speculation hiding downstream work, but candidate success is confounded with transcript stability and turn difficulty; it is not a causal estimate of an 846 ms speedup. Kyutai’s fixed first-frame computation remained the largest warm downstream cost.

Table 5: Deployment case-study measurements. Aggregate response latency covers three sessions; component medians come from the 13-turn fully exported trace and overlap in time.

Measurement	Observed value
Sessions / measured turns	3 / 36
Final VAD to first server audio, median	758 ms
Final VAD to first server audio, mean	932 ms
Final VAD to first server audio, range	528–1,652 ms
Turns below 800 ms	21 / 36
Fully traced turns	13
Final VAD to endpoint commitment, median	502 ms
Generation start to LLM first word, median	336 ms
TTS first word to first PCM, median	444 ms
Server audio send to browser render, median	95 ms
Final VAD to browser render, median	836 ms

The original 250–600 ms engineering target was not met consistently. The 800 ms line became a practical development reference, not a population target. The sessions were neither scripted nor blinded, involved one operator, and omit users, acoustic conditions, accents, and networks outside the development setting. They do not justify population p50 or

p95 claims or independently quantify interruption, duck, or backchannel-resume latency.

9 Observability and reproducibility

The browser’s diagnostic stream separates model evidence from durable conversation. Its rolling 20-second view records user-speech and assistant-audible lanes, raw adapter outputs, Silero state, applicability, causal audio time, age, inference latency, policy decisions, and browser acknowledgments. The adapter heads were not independently calibrated as probabilities of observable conversational behavior; their values are therefore retained as engineering telemetry rather than presented as behavioral results. Figure 4 limits the paper view to directly observable speech and playback activity.

Concrete reproducibility artifacts accompany each boundary: immutable serialized schemas reject extra fields; manifests retain seeds, input hashes, configuration, metrics, and checkpoint identity; locked evaluation files carry content hashes and detector provenance; and build and test commands are recorded with pinned model revisions. Deployment packages the final adapter and mounts persistent model caches without committing credentials. Table 6 identifies the final runtime snapshot and principal pins.

The public artifacts include the tool-use dataset, LoRA, merged 1.7B checkpoint, and synthetic turn-taking audio [27, 28, 29, 30]. Human conversational audio remains restricted by its source terms. The source repository contains generation, preparation, training, and evaluation code; narrative result summaries; the runtime and browser; and Modal deployment runbooks. Large prediction inventories and restricted source audio are not public artifacts.



Figure 4: Observable activity during a 20-second excerpt from an interactive microphone session. Green spans show detected user speech and gold spans show acknowledged assistant playback. The session exercised short backchannels and a floor-taking interruption, but the lanes intentionally do not assign semantic labels to individual overlaps.

10 Discussion and limitations

The principal result is not a learned endpointer that replaces silence timing. It is a bounded way to integrate uncertain learned evidence into a streaming controller. Immediate onset remains acoustic, early reactions are reversible, private work is discarded safely, and a deadline provides progress when the learned model is late or unavailable. This separation allowed the adapter to remain an optional signal after its standalone completion policy failed the locked gate.

The experiments also expose two transfer failures. Synthetic turn pretraining produced strong in-domain ranking but transferred weakly to human completion labels. Synthetic tool fine-tuning improved a shared-generator protocol holdout but did not establish general reasoning or retrieval quality. Qwen3-4B was selected for deployment after an engineering integration review, not a controlled model-quality comparison. Together, these results argue for measuring learned components at the boundary where they will be used rather than inferring deployment value from training-domain metrics.

Several limitations constrain the evidence. The locked V1 turn test contains 11 conversations and only 37 HOLD cases, so one error changes false cutoff by 2.70 percentage points and candidates within a conversation are correlated. Possible training-source overlap prevents a clean Smart Turn superiority comparison. V2 excludes 216 ambiguous cases and lacks the planned independent human label audit. Most importantly, the deployed step-750 adapter was not evaluated on the locked V1 test, and its floor-take and backchannel heads have not been calibrated against an independently annotated natural-conversation test set.

The deployment study adds only three unscripted sessions from one operator. It is not a user study and does not establish naturalness, interruption accuracy, or latency distributions across users, accents, noise, echo, networks, or GPU types. The system is English-first, human audio remains restricted by source terms, and factual generation depends on a small language model and external retrieval. A stronger evaluation would use conversation-disjoint human labels, participant-level confidence intervals, controlled overlap scenarios, and final-topology action latencies. A learned policy should replace the hybrid controller only if it improves the latency-versus-false-cancel frontier on that evidence.

11 Conclusion

VOICE-LIGHT demonstrates an artifact-backed path from causal conversational data to a public full-duplex cascaded agent [25, 26]. The system shares streaming ASR features with a small turn adapter, prepares responses speculatively, executes typed tools, controls playback reversibly, and records only acknowledged audio as durable assistant history. Its three-session case study shows sub-second median server response while retaining scale-to-zero deployment.

The negative result is equally central: the historical learned completion policy did not beat simple deployable timing baselines on the locked real-conversation test. Instrumentation therefore changed the design rather than being used to justify the original model. The resulting contribution is a reproducible hybrid system and a set of explicit evidence boundaries for future controlled turn-taking evaluation.

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A Reproducibility snapshot

Table 6: Principal revisions, hashes, and recorded validation for the final runtime. Full commands and additional hashes are recorded in the repository runbooks [26].

Artifact	Revision, hash, or recorded result
Report source	Committed TeX, figure assets, and build instructions in the source repository
Runtime source snapshot	e2f79adc on master; evaluated two-GPU deployment d78115d2
Qwen3-4B-Instruct-2507	cdbee75f17c01a7cc42f958dc650907174af0554
Qwen3-0.6B summarizer	c1899de289a04d12100db370d81485cdf75e47ca
Nemotron Streaming 0.6B	ebe59e5a817142986528bbbee5dba8db7b38ed50
Kyutai TTS 1.6B	f65439609986c392cb12df63938abcc550c3fb15
Deployed adapter step 750	SHA-256 d5c8e02c61dc9c230eac57992278383b81f14e3b71935b8dc684fe5da4171011
Release engineering checks	630 Python tests (1 skipped, 1 integration deselected), 38 browser tests, Ruff clean; not scientific evaluation

A.1 Build and validation commands

The release PDF is built from the committed source. From the repository root on Windows, the recorded validation sequence is:

```
.\.venv\Scripts\ruff.exe format --check --exclude .cache .
.\.venv\Scripts\ruff.exe check --exclude .cache .
.\.venv\Scripts\python.exe -m pytest tests\voice_agent tests\training\turn_taking
    -m "not integration" --import-mode=importlib
node --test tests\browser\*.test.mjs
tectonic --outdir output\pdf docs\technical-report\voice-light-technical-report.tex
```

Model-dependent integration tests and the deployed microphone acceptance session require the documented external services, GPU environment, and uncommitted credentials; they are not implied by the local command sequence above.